

Supplementary Information

A Unified Ornstein–Uhlenbeck–Lévy–Branching Framework for Interpretable Modeling of Cancer Evolution

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Supplementary Note 1. Complete Mathematical Formulation

The Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework represents biological evolution as a latent stochastic process that integrates continuous constrained dynamics, discontinuous evolutionary transitions, lineage-associated branch-state dynamics, environmental or therapy-dependent perturbations, and heterogeneous observation processes within a unified probabilistic architecture.

The framework is formulated at the level of a latent biological state. Experimental measurements are linked to this latent process through a separate observation model, allowing the same stochastic dynamics to be applied across longitudinal, spatial, bulk, single-cell, and multimodal biological datasets.

Latent biological state

Let

$$X_t \in \mathbb{R}^q$$

denote the latent biological state at continuous time t , where q is the dimension of the latent representation.

Depending on the biological application, X_t may represent

- pathway activities,
- ecological compositions,
- cell-state mixtures or abundances,
- developmental coordinates,
- spatial ecological states,
- molecular latent embeddings,
- or other learned low-dimensional biological representations.

The OULB framework intentionally separates latent biological dynamics from the experimental process through which those dynamics are observed.

Continuous Ornstein–Uhlenbeck dynamics

Conditional on the active branch state B_t , the latent state evolves according to a multivariate Ornstein–Uhlenbeck process,

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t,$$

where

- $W_t \in \mathbb{R}^q$ is a standard multivariate Wiener process;
- $\Theta_{B_t} \in \mathbb{R}^{q \times q}$ is the branch-specific restoring matrix;
- $\mu_{B_t}(t) \in \mathbb{R}^q$ is the time-dependent, branch-specific attractor;
- $\Sigma_{B_t} \in \mathbb{R}^{q \times q}$ is the branch-specific diffusion matrix.

For the general matrix-valued formulation, mean-reverting dynamics require the eigenvalues of Θ_b to have positive real parts. Under this sign convention, the deterministic component continuously drives X_t toward the branch-specific attractor $\mu_b(t)$.

The general OULB formulation permits full restoring and diffusion matrices. The benchmark simulation implementation used in this study employs diagonal restoring dynamics with nonnegative component-specific restoring parameters and diagonal diffusion with component-specific diffusion scales. This parameterization permits exact component-wise Ornstein–Uhlenbeck transitions.

Brownian diffusion baseline

Brownian motion with drift is treated as the corresponding unconstrained diffusion baseline,

$$dX_t = \beta dt + \Sigma dW_t,$$

where

$$\beta \in \mathbb{R}^q$$

denotes the drift vector.

When

$$\beta = 0,$$

the process reduces to driftless Brownian motion.

Brownian motion is therefore used as an unconstrained diffusion baseline rather than being defined solely through the restriction $\Theta = 0$ of the mean-reverting OULB equation.

Therapy-dependent attractor dynamics

External perturbations may modify the branch-specific evolutionary equilibrium through a time-dependent attractor,

$$\mu_{B_t}(t) = \mu_{0,B_t} + \Delta_{B_t} u(t),$$

where

- $\mu_{0,B_t} \in \mathbb{R}^q$ is the baseline or pre-treatment attractor for branch B_t ;

- $\Delta_{B_t} \in \mathbb{R}^q$ denotes the branch-specific attractor displacement induced by treatment or another external perturbation;
- $u(t)$ is the corresponding exposure or covariate function.

The exposure function may represent, for example,

- a step function indicating treatment onset,
- continuous treatment intensity,
- a pharmacokinetic exposure profile,
- developmental or environmental covariates,
- or another time-varying external signal.

More generally, the attractor may be represented as

$$\mu_{B_t}(t) = f_{B_t}(Z_t),$$

where Z_t denotes exogenous covariates and f_{B_t} may be linear or nonlinear. The additive treatment-shift formulation above is the primary specification used in the present implementation.

Compound-Poisson Lévy jump dynamics

Abrupt evolutionary transitions are represented through a compound-Poisson jump process,

$$J_t = \sum_{k=1}^{N_t} Z_k,$$

where

$$N_t \sim \text{Poisson}(\lambda t)$$

is a Poisson counting process with jump intensity

$$\lambda > 0,$$

and

$$Z_k \in \mathbb{R}^q$$

denotes the k th jump mark.

In the primary implementation,

$$Z_k \sim \mathcal{N}(0, s_J^2 I_q),$$

where $s_J > 0$ is the per-coordinate jump-scale parameter and I_q is the $q \times q$ identity matrix.

Jump marks are independently generated across events. The software also permits zero-centered Laplace-distributed jump marks as an alternative heavy-tailed specification.

The resulting process is a finite-activity compound-Poisson Lévy process. It represents rare discontinuous state changes while retaining continuous Ornstein–Uhlenbeck evolution between jump events.

Equivalently, a compound-Poisson Lévy process may be represented using a Poisson random measure,

$$J_t = \int_{[0,t] \times \mathbb{R}^q} z N(ds, dz),$$

with finite Lévy measure. The explicit compound-Poisson representation is used throughout the simulations and computational implementation in the present study.

Branch-state dynamics

Let

$$B_t \in \{1, \dots, K\}$$

denote the discrete branch state associated with the latent trajectory.

Branch-state evolution is modeled as a continuous-time Markov chain with generator matrix

$$Q = (q_{k\ell})_{k,\ell=1}^K,$$

where

$$q_{k\ell} \geq 0, \quad k \neq \ell,$$

and

$$q_{kk} = -\sum_{\ell \neq k} q_{k\ell}.$$

The finite-time transition matrix over an interval $\Delta > 0$ is

$$P(\Delta) = \exp(Q\Delta),$$

such that

$$P_{k\ell}(\Delta) = \Pr(B_{t+\Delta} = \ell \mid B_t = k).$$

Each branch may possess distinct dynamical parameters,

$$\Theta_k, \quad \mu_k(t), \quad \Sigma_k.$$

Branch-state transitions may represent lineage-associated state transitions, ecological niche switching, differentiation, or treatment-induced state conversion.

In the present implementation, the term *branching* refers to the organization of trajectories into lineage-associated evolutionary branches. Transitions among branch-specific dynamical regimes are governed by the continuous-time Markov chain B_t . An explicit birth–death branching process is not required by the core OULB implementation.

Unified latent-state dynamics

Combining continuous mean-reverting evolution, therapy-dependent attractors, branch-state dynamics, and discontinuous Lévy jumps gives

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t + dJ_t.$$

This expression defines the principal latent-state dynamics of the OULB framework.

Observation model

Observed measurements are linked to the latent process through a separate observation layer.

For observation i obtained at time t_i ,

$$Y_i = H_i X_{t_i} + \varepsilon_i,$$

with

$$Y_i \in \mathbb{R}^{p_i}, \quad H_i \in \mathbb{R}^{p_i \times q},$$

and

$$\varepsilon_i \sim \mathcal{N}(0, R_i), \quad R_i \in \mathbb{R}^{p_i \times p_i}.$$

Here,

- H_i is an observation or projection matrix linking the latent state to the measured variables;
- R_i is the observation-error covariance matrix.

Both H_i and R_i may vary across observations, allowing heterogeneous measurement processes across sampling times, experimental technologies, or data modalities.

The linear-Gaussian observation model above is the primary formulation used in the present study. More general observation likelihoods, including count, compositional, or other non-Gaussian measurement models, may be incorporated without changing the underlying latent OULB dynamics.

Probabilistic inference

Let

$$\Phi$$

denote the collection of model parameters, including, as applicable,

$$\Phi = \{\Theta, \mu, \Sigma, \Delta, Q, \lambda, s_J, H, R\}.$$

A general Bayesian posterior factorization may be written as

$$p(X, B, J, \Phi \mid Y) \propto p(Y \mid X, \Phi) p(X \mid B, J, \Phi) p(B \mid \Phi) p(J \mid \Phi) p(\Phi).$$

The observation likelihood

$$p(Y \mid X, \Phi)$$

is induced by the observation model, whereas

$$p(X \mid B, J, \Phi)$$

is induced by the continuous-time OULB latent dynamics.

The factors

$$p(B \mid \Phi)$$

and

$$p(J \mid \Phi)$$

describe the branch-state and jump processes, respectively.

The OULB architecture supports fully Bayesian inference for latent states and model parameters. In the validation experiments reported in this study, however, component-specific inferential procedures were used where computationally appropriate. Selected continuous-parameter recovery experiments used exact-transition maximum-likelihood estimation, whereas specialized procedures were used for jump detection and branch reconstruction. Fully Bayesian implementations may use Hamiltonian Monte Carlo or related posterior-sampling algorithms. Detailed inference procedures are provided in Supplementary Note 3.

Complete OULB model

The complete OULB framework can therefore be summarized as

$$\begin{aligned}
dX_t &= \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t + dJ_t, \\
\mu_{B_t}(t) &= \mu_{0,B_t} + \Delta_{B_t} u(t), \\
J_t &= \sum_{k=1}^{N_t} Z_k, \quad N_t \sim \text{Poisson}(\lambda t), \\
Z_k &\sim \mathcal{N}(0, s_J^2 I_q), \\
B_t &\sim \text{CTMC}(Q), \\
Y_i &= H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i).
\end{aligned}$$

This formulation separates the latent biological process from the observation process while retaining explicit representations of continuous constrained evolution, external perturbations, finite-activity Lévy jumps, lineage-associated branch-state dynamics, and measurement uncertainty within a common probabilistic framework.

Supplementary Note 2. Model Hierarchy and Special Cases

The OULB framework organizes established stochastic models as corresponding baselines, nested or constrained formulations, and progressively richer combinations of model components.

Brownian motion provides the unconstrained diffusion baseline. Classical Ornstein–Uhlenbeck (OU) dynamics introduce mean reversion toward a fixed attractor. Therapy-shifted OU models permit the attractor to vary in response to external perturbations. OU–Lévy models augment continuous mean-reverting dynamics with discontinuous compound-Poisson jump events, whereas OU–Branching formulations introduce lineage-associated branch-state switching. The complete OULB model combines all of these components within a common latent-state architecture.

The observation model is treated as a separate layer and may be applied to any of the latent dynamical formulations below.

Model hierarchy and special cases within the OULB framework.

Model	OU	Therapy-dependent attractor	Lévy jumps	Branch state
Brownian motion baseline	No	No	No	No
Classical OU	Yes	No	No	No
Treatment-shifted OU	Yes	Yes	No	No
OU–Lévy	Yes	Optional	Yes	No
OU–Branching	Yes	Optional	No	Yes
Full OULB	Yes	Yes	Yes	Yes

Brownian diffusion baseline

Brownian motion with drift is represented as

$$dX_t = \beta dt + \Sigma dW_t,$$

where $\beta \in \mathbb{R}^q$ denotes the drift vector and Σ controls diffusion.

This model represents unconstrained stochastic evolution without a restoring attractor, jump process, or branch-state dynamics.

Driftless Brownian motion is recovered when

$$\beta = 0.$$

Classical Ornstein–Uhlenbeck model

The classical OU model introduces mean reversion toward a fixed attractor,

$$dX_t = \Theta (\mu_0 - X_t) dt + \Sigma dW_t.$$

This formulation corresponds to continuous constrained stochastic dynamics without treatment-dependent attractor shifts, discontinuous jumps, or branch-state transitions.

Treatment-shifted Ornstein–Uhlenbeck model

Treatment-dependent dynamics are obtained by allowing the attractor to vary through time,

$$\mu(t) = \mu_0 + \Delta u(t),$$

giving

$$dX_t = \Theta [\mu_0 + \Delta u(t) - X_t] dt + \Sigma dW_t.$$

This formulation represents externally driven changes in evolutionary equilibrium while retaining continuous OU dynamics.

OU–Lévy model

Discontinuous evolutionary transitions are introduced by activating the compound-Poisson Lévy jump process,

$$dX_t = \Theta [\mu(t) - X_t] dt + \Sigma dW_t + dJ_t,$$

with

$$J_t = \sum_{k=1}^{N_t} Z_k, \quad N_t \sim \text{Poisson}(\lambda t).$$

In the present implementation,

$$Z_k \sim \mathcal{N}(0, s_J^2 I_q).$$

The attractor $\mu(t)$ may remain fixed or may include a therapy-dependent shift.

OU–Branching model

Lineage-associated branch-state dynamics are introduced through

$$B_t \in \{1, \dots, K\},$$

with B_t evolving according to a continuous-time Markov chain with generator matrix Q . Conditional on the active branch state,

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t.$$

Each branch may therefore possess distinct restoring dynamics, attractors, and diffusion scales.

The term *branching* refers to the organization of trajectories into lineage-associated evolutionary branches. The core implementation does not require an explicit birth–death branching process.

Full OULB model

The complete OULB formulation combines branch-specific OU dynamics, therapy-dependent attractor modulation, compound-Poisson Lévy jumps, and branch-state switching,

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t + dJ_t,$$

with

$$\mu_{B_t}(t) = \mu_{0,B_t} + \Delta_{B_t} u(t).$$

Observed measurements are linked to the latent process through the separate observation model,

$$Y_i = H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i).$$

Thus, increasing model complexity is achieved through progressive activation of interpretable dynamical components while preserving a common latent-state and observation architecture.

Supplementary Note 3. Bayesian Inference

The OULB framework supports probabilistic inference for latent states, continuous dynamical parameters, discontinuous jump events, branch-state dynamics, and observation uncertainty. Because the validation experiments examine distinct components of the framework under different simulation settings, inference was performed using component-specific procedures appropriate to each analysis rather than requiring a single computational algorithm for every experiment.

The general OULB architecture nevertheless admits a unified Bayesian formulation, as described below.

General posterior formulation

Let

$$Y = \{Y_i\}_{i=1}^n$$

denote the observed data,

$$X = \{X_{t_i}\}_{i=1}^n$$

the corresponding latent biological states,

$$B = \{B_t\}$$

the branch-state process,

and

$$J = \{J_t\}$$

the discontinuous jump process.

Let

$$\Phi$$

denote the collection of model parameters, including, as applicable,

$$\Phi = \{\Theta, \mu, \Sigma, \Delta, Q, \lambda, s_J, H, R\}.$$

A general posterior factorization may be written as

$$p(X, B, J, \Phi \mid Y) \propto p(Y \mid X, \Phi) p(X \mid B, J, \Phi) p(B \mid \Phi) p(J \mid \Phi) p(\Phi).$$

Here,

$$p(Y \mid X, \Phi)$$

is the observation likelihood,

$$p(X \mid B, J, \Phi)$$

is induced by the continuous-time OULB latent dynamics,

$$p(B \mid \Phi)$$

describes the branch-state process,

$$p(J \mid \Phi)$$

describes the compound-Poisson jump process,
and

$$p(\Phi)$$

contains the parameter priors.

Components absent from a particular constrained formulation are fixed or omitted from the corresponding posterior.

Observation likelihood

For the linear-Gaussian observation model used in the principal implementation,

$$Y_i = H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i),$$

the observation likelihood is

$$p(Y \mid X, \Phi) = \prod_{i=1}^n \mathcal{N}(Y_i \mid H_i X_{t_i}, R_i).$$

This formulation explicitly separates measurement uncertainty from the latent evolutionary process.

Alternative observation likelihoods may be substituted when required by a particular measurement technology without changing the underlying OULB dynamics.

Continuous-parameter estimation

Selected simulation benchmarking experiments used maximum-likelihood estimation to provide computationally efficient diagnostics of parameter recovery.

For a scalar Ornstein–Uhlenbeck process with constant parameters,

$$dX_t = \theta(\mu - X_t) dt + \sigma dW_t,$$

the exact transition distribution over an interval

$$\Delta_i = t_{i+1} - t_i$$

is

$$X_{t_{i+1}} \mid X_{t_i} \sim \mathcal{N}(m_i, v_i),$$

where

$$m_i = \mu + (X_{t_i} - \mu) e^{-\theta \Delta_i},$$

and, for $\theta > 0$,

$$v_i = \frac{\sigma^2}{2\theta} \left(1 - e^{-2\theta \Delta_i}\right).$$

The corresponding exact irregular-time log-likelihood is

$$\log L(\theta, \mu, \sigma) = \sum_{i=1}^{n-1} \log \mathcal{N}(X_{t_{i+1}} \mid m_i, v_i).$$

This exact-transition formulation avoids discretization error and naturally accommodates irregular observation intervals.

For the Brownian baseline,

$$dX_t = \beta dt + \sigma dW_t,$$

the transition distribution is

$$X_{t_{i+1}} \mid X_{t_i} \sim \mathcal{N}(X_{t_i} + \beta \Delta_i, \sigma^2 \Delta_i).$$

Treatment-shifted OU parameters were estimated analogously using the time-dependent attractor

$$\mu(t) = \mu_0 + \Delta u(t).$$

The benchmark implementation uses diagonal restoring and diffusion components, so multivariate trajectories can be evaluated through component-wise exact transition densities.

Measurement-aware OU inference

Observation-noise robustness experiments compared naive OU fitting with measurement-aware inference.

Under the observation model

$$Y_i = X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i),$$

measurement-aware inference explicitly incorporates observation uncertainty rather than attributing all observed variation to the latent diffusion process.

This distinction is particularly important under sparse, irregularly sampled, or noisy observation regimes, where naive fitting may inflate estimates of latent stochastic variation or drive parameters toward optimization boundaries.

Jump-event inference

Discontinuous-event validation used standardized innovations relative to the expected OU transition.

For each observation interval, the standardized innovation may be written schematically as

$$z_i = \frac{X_{t_{i+1}} - \mathbb{E}[X_{t_{i+1}} | X_{t_i}]}{\sqrt{\text{Var}[X_{t_{i+1}} | X_{t_i}]}}.$$

Large standardized innovations provide evidence for a discontinuous transition beyond that expected under continuous OU dynamics.

For the supplementary single-jump validation experiment, jump events were evaluated using a spike-and-slab-inspired event model that assigns interval-specific jump probabilities. The interval with the strongest jump evidence provides the estimated jump location, and the corresponding discontinuity is used to estimate jump size.

Performance was evaluated using

- jump-time localization error,
- true versus estimated jump size,
- receiver operating characteristic analysis,
- precision–recall analysis,
- detection thresholds, and
- false-positive rates under pure OU trajectories.

The main-text compound-Poisson stress tests additionally evaluated jump recovery across varying jump scales and observation-noise levels.

Branch-state inference

Branch reconstruction was evaluated using probabilistic mixture models applied to trajectory-level latent features.

For a model with K inferred branch classes, each trajectory may be associated with branch-membership probabilities

$$\gamma_{ik} = \Pr(B_i = k \mid Y_i), \quad k = 1, \dots, K.$$

Because mixture-model labels are exchangeable, inferred branch labels were optimally aligned to the true simulated labels before calculating reconstruction metrics.

Optimal label alignment was performed by maximizing agreement between true and inferred branch assignments.

Branch-state performance was summarized using

- overall branch-assignment accuracy,
- balanced accuracy,
- macro-averaged F1 score,
- Adjusted Rand Index,
- branch-membership calibration where applicable, and
- recovery of branch-transition structure where applicable.

These reconstruction experiments provide a complementary validation of lineage-associated branch structure to the continuous-time Markov branch-state formulation defined in Supplementary Note 1.

Bootstrap uncertainty quantification

Bootstrap resampling was used where appropriate to quantify uncertainty in aggregate validation metrics.

For a performance statistic

$$T = T(D)$$

computed from simulated dataset D , bootstrap replicates

$$T^{(1)}, \dots, T^{(B)}$$

were generated by resampling the corresponding simulation units with replacement.

Bootstrap distributions were used to construct confidence intervals for summary quantities such as parameter-recovery metrics, jump-detection performance, branch-reconstruction accuracy, and sensitivity-analysis results.

Bootstrap confidence intervals should be distinguished from Bayesian credible intervals, because the former quantify resampling variability rather than posterior uncertainty.

Fully Bayesian OULB inference

The general OULB formulation may alternatively be estimated jointly within a fully Bayesian framework.

For example, posterior sampling may target

$$p(X, B, J, \Phi \mid Y)$$

using Hamiltonian Monte Carlo for differentiable continuous parameters, together with suitable treatment of discrete branch and jump variables.

Depending on the parameterization, discrete latent variables may be marginalized, sampled using specialized algorithms, or represented through alternative continuous or mixture parameterizations.

Prior distributions for the principal continuous parameters are summarized in Supplementary Table S2.

Importantly, Hamiltonian Monte Carlo was not required for all simulation benchmarks reported in this study; it represents one implementation strategy for joint Bayesian inference under the general OULB architecture.

Posterior convergence diagnostics

For fully Bayesian implementations based on Hamiltonian Monte Carlo, posterior convergence may be assessed using complementary diagnostics, including

- the rank-normalized potential scale reduction statistic $\hat{R}$, with values close to 1 and preferably $\hat{R} < 1.01$;
- bulk and tail effective sample sizes;
- visual inspection of trace plots;
- assessment of divergent transitions and sampler diagnostics; and
- posterior predictive checks.

No single diagnostic is sufficient in isolation; convergence and posterior adequacy should be evaluated jointly.

Posterior predictive inference

For posterior samples

$$\Phi^{(s)}, \quad X^{(s)}, \quad B^{(s)}, \quad J^{(s)},$$

posterior predictive quantities may be generated from

$$p(Y_{\text{new}} | Y) = \int p(Y_{\text{new}} | X, B, J, \Phi) p(X, B, J, \Phi | Y) dX dB dJ d\Phi.$$

Posterior predictive distributions permit evaluation of

- future latent trajectories,
- future observations,
- uncertainty in evolutionary state,
- jump-event uncertainty,
- branch-state uncertainty, and
- predictive calibration.

Bayesian model comparison

When comparison among competing Bayesian formulations is required, predictive model adequacy may be assessed using criteria such as

- the widely applicable information criterion (WAIC);
- Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO); and
- Bayes factors when marginal likelihoods are well defined and can be estimated reliably.

Model-comparison procedures are optional components of the OULB architecture and were not required for every validation experiment.

Because the framework organizes models of progressively increasing complexity, comparisons should account for both predictive performance and biological interpretability rather than favoring additional model components solely on the basis of increased flexibility.

Relationship between inferential strategies

The inference strategy used in the present study can therefore be summarized at three complementary levels:

1. **Component-specific benchmarking:** exact-transition maximum-likelihood estimation and specialized jump- and branch-recovery procedures were used for efficient simulation-based validation;
2. **Measurement-aware inference:** explicit observation models were used to distinguish latent process variation from measurement uncertainty;

3. **General Bayesian OULB inference:** the complete framework supports joint posterior inference over latent states, branch states, jump processes, and model parameters when a fully Bayesian analysis is required.

This modular inferential strategy allows computational methods to be matched to the scientific objective while preserving a common probabilistic formulation across the nested OULB model hierarchy.

Supplementary Note 4. Observation Models

A central feature of the OULB framework is the explicit separation between latent biological dynamics and the experimental processes through which those dynamics are observed.

The latent stochastic process

$$X_t \in \mathbb{R}^q$$

describes the underlying biological state, whereas observed measurements

$$Y_i$$

are generated conditionally on the latent state at observation time t_i .

This separation permits the same latent evolutionary model to be used across experimental technologies with different sampling structures, measurement dimensions, and uncertainty characteristics.

Linear-Gaussian observation model

The primary observation model used in the present implementation is

$$Y_i = H_i X_{t_i} + \varepsilon_i,$$

where

$$Y_i \in \mathbb{R}^{p_i}, \quad H_i \in \mathbb{R}^{p_i \times q},$$

and

$$\varepsilon_i \sim \mathcal{N}(0, R_i), \quad R_i \in \mathbb{R}^{p_i \times p_i}.$$

Here,

- X_{t_i} is the latent biological state at observation time t_i ;
- Y_i is the corresponding experimental observation;
- H_i is the observation or projection matrix linking the latent state to the measured variables;
- R_i is the observation-error covariance matrix;
- ε_i represents measurement uncertainty that is distinct from the stochastic variation of the latent biological process.

Conditional on the latent state,

$$Y_i \mid X_{t_i} \sim \mathcal{N}(H_i X_{t_i}, R_i).$$

The corresponding observation likelihood is

$$p(Y \mid X, H, R) = \prod_{i=1}^n \mathcal{N}(Y_i \mid H_i X_{t_i}, R_i).$$

Identity observation model

When the observed variables are represented in the same latent coordinate system as the underlying stochastic state,

$$p_i = q$$

and the observation matrix may be taken as

$$H_i = I_q,$$

giving

$$Y_i = X_{t_i} + \varepsilon_i.$$

Under isotropic Gaussian measurement noise,

$$R_i = \sigma_{\text{obs},i}^2 I_q,$$

so that

$$Y_i \mid X_{t_i} \sim \mathcal{N}(X_{t_i}, \sigma_{\text{obs},i}^2 I_q).$$

This formulation provides a direct representation of noisy measurements of the latent state and is particularly useful for simulation-based benchmarking.

Process variation and measurement uncertainty

The OULB framework distinguishes two conceptually different sources of stochastic variation.

The latent diffusion term

$$\Sigma_{B_t} dW_t$$

represents stochastic variation in the underlying biological process.

By contrast,

$$\varepsilon_i \sim \mathcal{N}(0, R_i)$$

represents uncertainty introduced during experimental observation.
Thus,

$$\text{biological process variation} \neq \text{measurement uncertainty}.$$

Failure to distinguish these quantities may cause experimental noise to be incorrectly attributed to biological diffusion or other latent dynamical parameters.

This distinction motivates the measurement-aware inference experiments reported in the main manuscript, in which explicit modeling of observation uncertainty was compared with naive fitting that does not separately account for measurement error.

Heteroscedastic observation uncertainty

The observation covariance matrix may vary among measurements,

$$R_i \neq R_j, \quad i \neq j,$$

allowing the framework to represent heteroscedastic measurement uncertainty.
For example,

$$R_i = \sigma_{\text{obs},i}^2 I_q$$

permits each observation to have its own measurement-error variance.
More generally, a non-diagonal covariance matrix

$$R_i$$

may represent correlated measurement errors among observed variables.
This flexibility permits uncertainty to vary across

- sampling times,
- patients or biological specimens,
- experimental batches,
- spatial locations,
- measurement technologies,
- or latent-state dimensions.

Irregular sampling

The latent OULB process evolves in continuous time and therefore does not require equally spaced observations.

For observation times

$$t_1 < t_2 < \dots < t_n,$$

define

$$\Delta_i = t_{i+1} - t_i.$$

The intervals

$$\Delta_i$$

may vary arbitrarily across observations.

For the OU component, the exact transition distribution depends explicitly on Δ_i , allowing irregular observation schedules to be accommodated without interpolating the latent trajectory onto an artificial regular grid.

Irregular sampling therefore affects the amount of information available about the latent process but does not alter the underlying continuous-time OULB formulation.

Missing observations

Missing observations are represented by the absence of a measurement likelihood contribution at the corresponding observation time or dimension.

If an observation at time t_i is unavailable, the latent process continues to evolve according to the OULB dynamics,

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t + dJ_t,$$

without requiring an artificial observed value.

Inference across the unobserved interval is therefore governed by the latent transition model and the observations available before and after the missing measurement.

This formulation distinguishes missing experimental measurements from interruptions in the underlying biological process.

Modality-specific projections

The observation matrix

$$H_i$$

provides a general interface between a shared latent biological state and modality-specific measurements.

For example, the latent state may represent a common biological coordinate system while observations are derived from

- longitudinal single-cell RNA sequencing,
- spatial transcriptomics,
- bulk molecular measurements,
- pathway-activity estimates,
- ecological-state compositions,
- imaging-derived latent representations,
- or multimodal molecular measurements.

Different experimental modalities may therefore use different observation matrices,

$$H_i,$$

and different observation covariance matrices,

$$R_i,$$

without changing the latent OULB dynamics.

This separation provides the basis for the technology-agnostic interpretation of the framework.

Latent-state preprocessing and observation modeling

The OULB framework begins after modality-specific preprocessing and latent-state construction.

For high-dimensional molecular measurements, preprocessing may include

- quality control,
- normalization,
- feature selection,
- dimensionality reduction,
- ecological or pathway scoring,
- latent representation learning,
- or projection into a common biological state space.

The resulting low-dimensional measurements are treated as observations of the latent stochastic process.

Accordingly, the OULB framework does not require a particular dimensionality-reduction algorithm, embedding method, or preprocessing pipeline.

Uncertainty introduced by preprocessing or measurement may be represented, when estimable, through the observation model rather than being incorporated into the latent biological diffusion by default.

Alternative observation likelihoods

The linear-Gaussian observation model is the primary formulation used in the present study.

The general separation between latent dynamics and observations, however, permits alternative likelihoods to be incorporated when required by the measurement scale.

Schematically,

$$Y_i \sim p_{\text{obs}}(Y_i \mid X_{t_i}, \phi_i),$$

where

$$\phi_i$$

denotes observation-model parameters.

Possible extensions include

- Poisson or negative-binomial likelihoods for count-valued measurements;
- binomial or beta-binomial likelihoods for proportion or success-count data;
- multinomial likelihoods for discrete compositional measurements;
- Dirichlet or logistic-normal models for continuous compositional representations;
- nonlinear observation mappings for complex multimodal measurements.

These likelihoods represent extensions of the general OULB architecture and were not required for the principal validation experiments reported in the present study.

Importantly, changing the observation likelihood does not require changing the latent evolutionary equation.

Measurement-aware inference

Under a measurement-aware formulation, inference is based jointly on the latent transition model and the observation likelihood,

$$p(X, \Phi \mid Y) \propto p(Y \mid X, \Phi) p(X \mid \Phi) p(\Phi),$$

or, for the complete OULB model,

$$p(X, B, J, \Phi \mid Y) \propto p(Y \mid X, \Phi) p(X \mid B, J, \Phi) p(B \mid \Phi) p(J \mid \Phi) p(\Phi).$$

Explicit observation modeling allows uncertainty in Y_i to be propagated into inference of latent states and dynamical parameters.

By contrast, naive fitting effectively treats the observed values as though they were exact realizations of the latent state,

$$Y_i \equiv X_{t_i},$$

thereby ignoring measurement uncertainty.

The stress-test experiments in the present study compare these two approaches under varying observation number, measurement noise, irregular sampling, and missing observations.

General observation architecture

The observation layer may therefore be summarized as

$$\begin{array}{l} X_t \sim \text{OULB latent dynamics,} \\ Y_i \mid X_{t_i} \sim p_{\text{obs}}(Y_i \mid X_{t_i}, \phi_i). \end{array}$$

For the primary linear-Gaussian implementation,

$$Y_i = H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i).$$

This architecture separates biological evolution from experimental measurement while permitting heterogeneous observation uncertainty, irregular sampling, missing measurements, and modality-specific projections within a common probabilistic framework.

Supplementary Note 5. Cross-scale Interpretation

The OULB framework is formulated in terms of a continuous latent biological state

$$X_t \in \mathbb{R}^q,$$

rather than a particular experimental measurement or biological scale. Consequently, the interpretation of X_t depends on the biological question and the latent representation constructed from the underlying data.

Possible levels of biological organization include

Genes $\rightarrow$ Pathways $\rightarrow$ Cell States $\rightarrow$ Ecological States $\rightarrow$ Spatial Niches $\rightarrow$ Patient Evolution.

At the molecular level, latent coordinates may summarize gene-expression programs, pathway activities, or other molecular features. At the cellular level, they may represent cell-state mixtures, differentiation states, or phenotypic programs. At higher levels of organization, the same latent-state formulation may represent ecological compositions, spatial niches, or longitudinal patient-level evolutionary states.

The arrows above indicate increasing levels of biological organization rather than a required deterministic mapping among scales. The OULB framework does not assume that every application contains all of these levels or that one scale can be uniquely reconstructed from another.

From the probabilistic perspective, the same latent dynamical equation

$$dX_t = \Theta_{B_t} [\mu_{B_t}(t) - X_t] dt + \Sigma_{B_t} dW_t + dJ_t$$

may be applied after an appropriate latent representation has been defined.

The biological interpretation of the model parameters therefore depends on the scale represented by X_t . For example,

- $\mu_{B_t}(t)$ represents the branch-specific equilibrium or attractor at the chosen biological scale;
- Θ_{B_t} describes the strength and timescale of restoration toward that attractor;
- Σ_{B_t} represents continuous stochastic variability around the constrained trajectory;
- J_t represents rare discontinuous changes in the latent state;
- B_t represents transitions among lineage-associated or ecologically distinct dynamical regimes.

Thus, OULB treats different latent biological representations using a common probabilistic architecture while preserving scale-specific biological interpretation.

Importantly, this cross-scale formulation does not imply that parameters estimated at different biological scales are numerically interchangeable. Rather, it provides a shared mathematical language through which continuous dynamics, discontinuous transitions, branch-state structure, and observation uncertainty can be interpreted across different levels of biological organization.

Supplementary Note 6. Simulation Framework

Simulation studies were used to evaluate the principal dynamical components of the OULB framework under controlled conditions in which the true generating parameters and latent states were known.

The validation architecture included

1. **Brownian diffusion baseline:** unconstrained stochastic trajectories with drift and diffusion;
2. **Classical Ornstein–Uhlenbeck dynamics:** continuous mean-reverting trajectories with fixed attractors;
3. **Treatment-shifted Ornstein–Uhlenbeck dynamics:** continuous trajectories with externally induced changes in the attractor;
4. **OU–Lévy dynamics:** mean-reverting trajectories augmented by discontinuous jump events;
5. **OU with branch-state dynamics:** trajectories evolving among branch-specific dynamical regimes;
6. **Combined OULB dynamics:** integrated simulations containing multiple continuous and discrete model components.

Continuous-process simulation

Brownian trajectories were generated according to

$$dX_t = \beta dt + \Sigma dW_t,$$

whereas OU trajectories followed

$$dX_t = \Theta (\mu - X_t) dt + \Sigma dW_t.$$

The benchmark implementation used diagonal restoring and diffusion components, permitting exact component-wise transition simulation.

Treatment-dependent simulations used

$$\mu(t) = \mu_0 + \Delta u(t),$$

where $u(t)$ specifies the treatment-exposure function.

Jump-process simulation

Discontinuous-event simulations used the compound-Poisson process

$$J_t = \sum_{k=1}^{N_t} Z_k, \quad N_t \sim \text{Poisson}(\lambda t),$$

with primary jump marks

$$Z_k \sim \mathcal{N}(0, s_J^2 I_q).$$

Simulation experiments varied the characteristic jump scale and observation noise to determine the operating regimes under which discontinuous events could be reliably detected.

Specialized supplementary experiments additionally evaluated localization and size recovery for controlled single-jump trajectories.

Branch-state simulation

Branch-state evolution was represented through a continuous-time Markov chain

$$B_t \in \{1, \dots, K\}$$

with generator matrix

$$Q.$$

For an interval of duration Δ , the branch-transition probabilities are given by

$$P(\Delta) = \exp(Q\Delta).$$

Conditional on the active branch state, trajectories evolve according to branch-specific OU parameters.

Simulation experiments varied the separation among branch-specific attractors and observation uncertainty to evaluate recovery of latent branch assignments.

Separate supplementary lineage-reconstruction experiments used controlled descendant-branch scenarios to evaluate probabilistic branch assignment and label reconstruction. These experiments complement the continuous-time branch-state formulation and should be interpreted within their corresponding simulation settings.

Observation-process simulation

Latent trajectories were projected through an observation model before inference.

The principal simulation observation model was

$$Y_i = H_i X_{t_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, R_i).$$

Stress-test experiments systematically varied

- number of observations,
- observation noise,

- regular versus irregular sampling,
- missing observations.

These experiments were used to compare measurement-aware inference with naive fitting that does not explicitly separate observation uncertainty from latent process variation.

Reproducibility

Each simulation condition consisted of repeated independent replicates. Random seeds and parameter configurations were specified through version-controlled configuration files and archived with the corresponding validation outputs.

Simulation scenarios and activated model components are summarized in Supplementary Table S3, with parameter settings and replicate configurations provided in Supplementary Data S1.

The supplementary validation experiments were designed to examine specific components of the framework and may therefore use parameter regimes distinct from those shown in the main-figure stress tests. Numerical performance should be interpreted within the simulation settings associated with each experiment.

This supplementary lineage-reconstruction benchmark represents a discrete descendant-branch assignment experiment and is complementary to the continuous-time branch-state switching formulation used in the general OULB model.

Supplementary Note 7. Parameter Recovery

Simulation-based validation was used to determine whether continuous model parameters and discrete latent structures could be recovered when the true generating values were known.

The principal recovery analyses evaluated

- Brownian drift and diffusion;
- OU attractor location;
- OU mean-reversion strength;
- OU diffusion scale;
- treatment-induced attractor displacement;
- jump-event detection, localization, and size or scale recovery, depending on the simulation experiment;
- latent branch-state assignment and branch-structure reconstruction.

Continuous-parameter recovery

For a generating parameter θ_0 and replicate estimates

$$\hat{\theta}_1, \dots, \hat{\theta}_n,$$

the mean estimate is

$$\bar{\hat{\theta}} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_i.$$

Estimation bias is defined as

$$\text{Bias} = \bar{\hat{\theta}} - \theta_0,$$

and root-mean-square error as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\theta}_i - \theta_0)^2}.$$

Sampling distributions across simulation replicates were additionally examined to characterize parameter-specific estimation variability.

Where interval estimates were generated in a particular supplementary experiment, empirical interval coverage was calculated as the fraction of simulation replicates for which the corresponding interval contained the true generating parameter.

The term *interval coverage* is used here unless the interval is specifically derived from a Bayesian posterior, in which case it may be described as credible-interval coverage.

Jump-event recovery

Jump-event recovery was evaluated using quantities appropriate to the corresponding validation experiment.

For binary jump detection, performance measures included

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

and

$$\text{FPR} = \frac{FP}{FP + TN}.$$

Supplementary single-jump experiments additionally evaluated

- true and detected jump times,
- absolute jump-time error,
- true and estimated jump sizes,
- jump-probability scores,
- receiver operating characteristic performance,
- precision–recall performance.

The main compound-Poisson stress tests evaluated recovery as a function of the configured jump-scale parameter s_J and observation noise.

Thus, *jump size* refers to the realized or deliberately injected discontinuity in experiments where an individual event is evaluated, whereas *jump scale* refers to the generative distributional parameter controlling compound-Poisson jump marks.

Branch-state recovery

Latent branch assignments were compared with their known simulated states.

Principal recovery metrics included

- branch-assignment accuracy,
- Adjusted Rand Index,

- branch-transition recovery where applicable.

Specialized supplementary branch-reconstruction experiments additionally evaluated quantities such as

- balanced accuracy,
- macro-averaged F1 score,
- branch-membership probabilities,
- Brier score,
- reconstructed branch-transition structure.

Because mixture-model labels are exchangeable, inferred branch labels were optimally aligned to the true simulated labels before calculating label-dependent performance measures.

Interpretation of recovery analyses

Parameter-recovery results were interpreted in terms of practical recoverability under the evaluated simulation conditions rather than as formal proofs of structural identifiability.

In particular, recovery performance depends on

- trajectory length,
- observation density,
- measurement uncertainty,
- jump scale,
- branch separation,
- and the strength of the underlying dynamical signal.

Accordingly, the validation experiments characterize operating regimes in which individual OULB components can be reliably recovered and identify conditions under which estimation becomes more difficult.

Supplementary Note 8. Sensitivity Analyses

Sensitivity analyses evaluated the robustness of OULB inference and prediction to changes in observation architecture, latent-state complexity, and computational scale.

The principal perturbations included

- observation noise;
- irregular sampling intervals;
- missing observations;
- sample size or observation number;
- latent-state dimensionality;
- jump scale;
- branch separation;
- computational runtime and scaling.

Observation noise

Increasing observation noise reduces the signal-to-noise ratio available for inference of continuous parameters, discontinuous events, and latent branch states.

Sensitivity experiments therefore evaluated recovery across progressively increasing measurement uncertainty.

Irregular sampling

Irregular observation schedules were generated by varying the intervals

$$\Delta_i = t_{i+1} - t_i.$$

Because OULB is formulated in continuous time, irregular sampling does not require interpolation to a regular time grid. Nevertheless, increasingly heterogeneous observation intervals may reduce the information available for parameter and event recovery.

Missing observations

Missing-data experiments removed a fraction of observations while allowing the latent process to evolve continuously through the unobserved intervals.

These analyses evaluated the degree to which parameter estimation, event detection, branch reconstruction, and prediction remained stable as observational completeness decreased.

Sample size and observation density

Sensitivity to data availability was evaluated by varying the number of observations or simulation units available for inference.

These experiments characterize how recovery and predictive performance change as the amount of observed information increases or decreases.

Latent-state dimensionality

The dimension

$$q$$

of the latent biological state was varied to evaluate computational and statistical behavior as the representation became increasingly multivariate.

These analyses assess whether performance remains stable as additional latent dimensions are introduced and quantify the associated computational cost.

Jump scale and branch separation

Discrete-event recovery depends on the strength of the latent signal relative to observation uncertainty.

Jump-detection analyses therefore varied the characteristic compound-Poisson jump scale

$$s_J,$$

whereas branch-recovery analyses varied the separation among branch-specific latent states or attractors.

These perturbations define practical signal-to-noise regimes for recovery of discontinuous events and branch-state structure.

Computational scaling

Computational runtime was evaluated as a function of increasing problem size, including sample size and latent-state dimensionality where applicable.

Runtime analyses were intended to characterize the computational behavior of the implemented workflow rather than establish algorithm-independent complexity bounds.

Interpretation of sensitivity experiments

The sensitivity analyses were designed as empirical robustness tests. Individual supplementary experiments may use parameter regimes distinct from those used in the main-text stress tests.

Consequently, apparently different absolute performance levels across figures or tables do not imply contradictory behavior; each result should be interpreted relative to its generating simulation regime.

Detailed sensitivity settings and replicate-level results are provided in Supplementary Table S8 and the corresponding Supplementary Data files.

Supplementary Note 9. Computational Implementation

The OULB framework was implemented as a modular Python computational workflow designed to separate simulation, inference, validation, calibration, and visualization.

Major computational stages include

1. preprocessing and latent-state construction,
2. latent-state projection,
3. stochastic parameter estimation,
4. jump-event detection,
5. branch-state reconstruction,
6. stochastic simulation,
7. benchmarking and validation,
8. empirical calibration,
9. visualization and figure generation.

Software environment

The computational workflow uses open-source Python scientific-computing libraries, including, where applicable,

- NumPy,
- SciPy,
- pandas,
- scikit-learn,
- PyMC,
- ArviZ,
- Matplotlib,
- Scanpy.

NumPy and SciPy provide numerical and stochastic-process functionality; pandas supports structured data manipulation; scikit-learn supports selected statistical-learning and mixture-model procedures; PyMC and ArviZ support Bayesian implementations and posterior diagnostics; Matplotlib supports visualization; and Scanpy is used where appropriate in preprocessing of single-cell data.

Not every software dependency is required for every OULB model component or validation experiment.

Exact-transition simulation

For the benchmark implementation, Brownian and diagonal OU processes are simulated using their exact finite-time transition distributions rather than Euler–Maruyama discretization.

For a scalar OU component,

$$X_{t+\Delta} \mid X_t \sim \mathcal{N} \left(\mu + (X_t - \mu)e^{-\theta\Delta}, \frac{\sigma^2}{2\theta} (1 - e^{-2\theta\Delta}) \right),$$

for

$$\theta > 0.$$

The $\theta = 0$ limit is treated as unconstrained diffusion where appropriate.

This exact-transition approach avoids numerical discretization error for the linear Gaussian continuous component.

Branch-state propagation

For a continuous-time branch-state generator

$$Q,$$

the finite-time transition matrix is computed as

$$P(\Delta) = \exp(Q\Delta),$$

using matrix exponentiation.

The branch state for the subsequent interval is sampled from the row of $P(\Delta)$ corresponding to the currently active state.

Conditional on the resulting branch state $B_t = b$, the trajectory is propagated using the corresponding branch-specific parameters

$$\Theta_b, \quad \mu_b(t), \quad \Sigma_b.$$

This implementation permits branch transitions to occur over irregular time intervals while preserving the continuous-time Markov formulation.

Compound-Poisson jump simulation

Discontinuous events are simulated through the compound-Poisson process

$$J_t = \sum_{k=1}^{N_t} Z_k,$$

where, for an interval of duration Δ ,

$$N_{\Delta} \sim \text{Poisson}(\lambda\Delta).$$

Conditional on

$$N_{\Delta} = n,$$

the n jump marks are generated independently.

For the primary Gaussian jump specification,

$$Z_k \sim \mathcal{N}(0, s_J^2 I_q).$$

The total jump displacement accumulated during the interval is therefore

$$\Delta J = \sum_{k=1}^n Z_k.$$

The software additionally permits independent zero-centered Laplace jump marks with scale parameter s_J .

Jump events and branch transitions are recorded in an event ledger that preserves the corresponding simulation interval and event-specific metadata for downstream validation.

Treatment-dependent dynamics

Treatment-dependent simulations modify the latent attractor through

$$\mu_b(t) = \mu_{0,b} + \Delta_b u(t).$$

The exposure function $u(t)$ is evaluated at the relevant simulation time and may represent a step change at treatment onset or another prespecified time-dependent perturbation.

Because treatment modifies the attractor rather than replacing the underlying stochastic process, the same transition architecture can be used before and after treatment while allowing the equilibrium state to change through time.

Observation simulation

Latent trajectories are converted to observed trajectories through a separate observation layer.

For the principal Gaussian implementation,

$$Y_i = H_i X_{t_i} + \varepsilon_i,$$

where

$$\varepsilon_i \sim \mathcal{N}(0, R_i).$$

Simulation utilities permit variation in

- observation-noise magnitude,
- observation schedule,
- irregular sampling,
- missing-observation probability,
- and whether latent and observed states are retained jointly.

This separation permits identical latent trajectories to be evaluated under different observation architectures without changing their underlying biological dynamics.

Component-specific inference

The computational workflow uses inference procedures matched to the corresponding validation objective.

Continuous Brownian and OU parameters are estimated using exact-transition likelihoods.

Selected measurement-aware OU analyses explicitly incorporate the observation-noise scale during parameter estimation.

Jump detection is based on deviations from the expected continuous OU transition and, depending on the validation experiment, may use standardized innovations, probabilistic jump scores, or threshold-based classification.

Branch reconstruction uses either the known simulated branch-state trajectory for direct recovery assessment or probabilistic trajectory-level clustering followed by optimal label alignment in specialized supplementary experiments.

These procedures are described in greater detail in Supplementary Note 3.

Benchmarking and validation

Benchmarking utilities calculate quantitative recovery metrics from known simulated truth and inferred quantities.

Continuous-parameter evaluation includes

- estimation bias,
- root-mean-square error,
- sampling distributions,
- and interval coverage where available.

Jump-event evaluation includes

- precision,
- recall,
- false-positive rate,
- jump-time localization error,
- and jump-size recovery where applicable.

Branch-state evaluation includes

- branch-assignment accuracy,
- Adjusted Rand Index,
- balanced accuracy and macro-F1 in selected supplementary analyses,
- and branch-transition recovery where applicable.

Sensitivity and robustness analyses additionally summarize predictive performance and numerical stability across perturbed simulation conditions.

Empirical calibration

The computational workflow includes a calibration layer that extracts summary quantities from processed longitudinal biological data and uses them to define biologically grounded simulation settings.

Calibration quantities may include

- observation counts,
- sampling intervals,
- empirical transition frequencies,
- latent-state displacement distributions,
- branch-switch frequencies,
- characteristic jump scales,
- and estimated observation uncertainty.

Calibration quantities are classified according to whether they are

- empirically measured,
- derived from empirical measurements,

- or prespecified when they are not identifiable from the available observations.

This distinction preserves provenance and avoids presenting non-identifiable simulation parameters as direct empirical estimates.

Configuration and deterministic reproducibility

Simulation and benchmarking settings are controlled through version-controlled configuration files.

Configuration parameters include, where applicable,

- model type,
- number of simulation replicates,
- random seed,
- observation number,
- follow-up duration,
- OU parameters,
- jump-process parameters,
- branch-transition parameters,
- observation-noise settings,
- and sensitivity-analysis conditions.

Deterministic random seeds are used to ensure reproducible simulation outputs.

Where multiple replicates are required, replicate-specific seeds are generated deterministically from the master configuration seed.

Archived intermediate outputs

Intermediate computational products are preserved rather than retaining only final figures.

Archived outputs include, where applicable,

- simulated latent trajectories,
- observed trajectories,
- parameter estimates,
- parameter-recovery summaries,
- jump-event classifications,

- jump-detection metrics,
- branch assignments,
- branch-recovery metrics,
- calibration summaries,
- sensitivity-analysis outputs,
- predictive evaluations,
- and figure-ready summary tables.

Preserving these intermediate products allows each reported figure and table to be traced back to the underlying simulation or inference output.

Software provenance and availability

The complete source code, configuration files, simulation scripts, benchmarking workflows, validation outputs, and figure-generation scripts are maintained in the accompanying version-controlled repository.

The software associated with the manuscript is publicly available at

https://github.com/shkim9391/Unified_OU_Levy_Branching_Framework

and the archived release corresponding to the study is permanently available through Zenodo:

<https://doi.org/10.5281/zenodo.21827910>.

The archived release provides a fixed computational reference for the analyses reported in the manuscript, while the version-controlled repository permits future methodological development.

Computational workflow summary

The computational implementation may therefore be summarized as

Configuration → Simulation or empirical input

→ Latent-state representation

→ Observation model

→ Parameter/event/branch inference

→ Benchmarking and calibration

→ Archived outputs

→ Figures and tables.

This modular architecture separates individual analytical stages while maintaining reproducibility and provenance across the complete OULB workflow.

Supplementary Note 10. Illustrative Leukemia Application

The pediatric leukemia analyses provide an empirical reference for the general OULB framework rather than serving as an independent disease-specific validation study.

The principal purpose of the leukemia application is to demonstrate how empirical biological observations can inform latent-state construction, observation architecture, and simulation calibration while remaining separate from the general mathematical formulation.

Public datasets

Publicly available datasets used for empirical calibration or as reference examples include

- GEO accession GSE235063, containing longitudinal single-cell RNA-sequencing measurements from pediatric leukemia;
- GEO accession GSE253355, providing normal bone-marrow single-cell reference data;
- GEO accession GSE279576, providing spatial transcriptomic bone-marrow specimens;
- GEO accession GSE235923, containing longitudinal pediatric acute myeloid leukemia measurements.

These datasets were used either to derive empirical calibration quantities or to illustrate how modality-specific latent representations may interface with the OULB framework.

No new experimental samples were generated for the present study.

Latent-state construction

Raw or processed molecular measurements were transformed using modality-appropriate preprocessing procedures before being supplied to the stochastic framework.

Depending on the dataset, preprocessing may include

- quality control,
- normalization,
- dimensionality reduction,
- pathway or ecological scoring,
- latent-state construction,
- and projection into low-dimensional biological coordinates.

The resulting coordinates are treated as observations of an underlying latent state,

$$X_t \in \mathbb{R}^q.$$

The OULB framework begins after this latent-state representation has been defined and therefore does not depend on a specific embedding, dimensionality-reduction, or molecular preprocessing algorithm.

Empirical observation architecture

Longitudinal leukemia measurements provide an empirical basis for characterizing realistic observation structures.

Quantities extracted from the empirical data include, where available,

- number of observations per patient,
- time intervals between observations,
- empirical transition frequencies,
- latent-state displacement distributions,
- branch-switch frequencies,
- characteristic discontinuous displacement scales,
- and observation-noise estimates.

These quantities define the observational and dynamical scales used in empirically calibrated simulations.

Calibration provenance

Not every OULB parameter can be directly identified from sparse longitudinal observations.

Calibration quantities are therefore classified as

1. **Empirically measured:** quantities directly calculated from observed longitudinal data;
2. **Derived:** quantities calculated from empirical measurements through a defined summary or transformation;
3. **Prespecified:** quantities assigned for simulation because they cannot be uniquely estimated from the available data.

This classification prevents simulated mechanistic parameters from being misinterpreted as direct biological estimates.

Empirically calibrated simulations

Empirically calibrated OULB simulations use the observed data architecture to construct synthetic trajectories with realistic sampling density and characteristic dynamical scales.

The purpose of calibration is not to reproduce every empirical statistic exactly. Instead, calibration provides a biologically grounded operating regime in which the methodological behavior of the stochastic framework can be evaluated.

Calibrated trajectories may exhibit qualitatively distinct dynamical patterns such as

- retention-dominated evolution,
- discontinuous or jump-dominated evolution,
- branch-state reorganization,
- and combinations of continuous and discrete evolutionary changes.

Observed and simulated summary statistics are compared to evaluate the degree to which the calibrated simulations reproduce the empirical observation architecture and characteristic dynamical scale.

Interpretational scope

The leukemia application is illustrative.

The framework does not imply that every observed latent displacement corresponds uniquely to a specific molecular mechanism, nor that every inferred branch transition corresponds to a literal cellular birth–death branching event.

Instead, the latent-state representation provides a statistical description of longitudinal biological evolution at the scale defined by the selected features or embedding.

Accordingly,

- continuous OU dynamics represent constrained evolution around an effective attractor;
- therapy-dependent attractor shifts represent changes in the effective evolutionary landscape;
- compound-Poisson jumps represent rare discontinuous changes not adequately described by continuous Gaussian diffusion;
- branch states represent lineage-associated or ecologically distinct dynamical regimes;
- the observation model represents uncertainty associated with sparse and noisy experimental measurement.

These quantities should therefore be interpreted as components of an explicit probabilistic model rather than as one-to-one surrogates for individual molecular events.

Relationship to methodological validation

The empirical leukemia application complements, but does not replace, simulation-based validation. Simulation experiments provide known ground truth for evaluating

- parameter recovery,
- jump-event detection,
- branch-state reconstruction,
- observation-model robustness,
- and computational sensitivity.

The empirical data, by contrast, provide realistic observation architectures and characteristic scales for calibration.

Together, these two components establish a bridge between controlled methodological benchmarking and biologically plausible application.

Generalization beyond leukemia

The OULB framework is not specific to leukemia or cancer.

Because the mathematical architecture operates on a generic latent state

$$X_t \in \mathbb{R}^q,$$

the same framework may be applied to biological systems exhibiting continuous constrained evolution, rare discontinuous transitions, state diversification, environmental perturbation, and imperfect observation.

Potential applications include

- developmental lineage specification,
- immune-cell differentiation,
- microbial evolution,
- ecological community dynamics,
- tissue regeneration,
- adaptive evolution,
- host–pathogen interactions,
- and phylogenetic or comparative biological systems.

The biological interpretation of the latent state and model parameters will differ among applications, but the underlying probabilistic architecture remains unchanged.

Summary

The illustrative leukemia application demonstrates how empirical longitudinal and multimodal biological data can be connected to the general OULB architecture without embedding disease-specific assumptions into the stochastic model itself.

The relationship may be summarized as

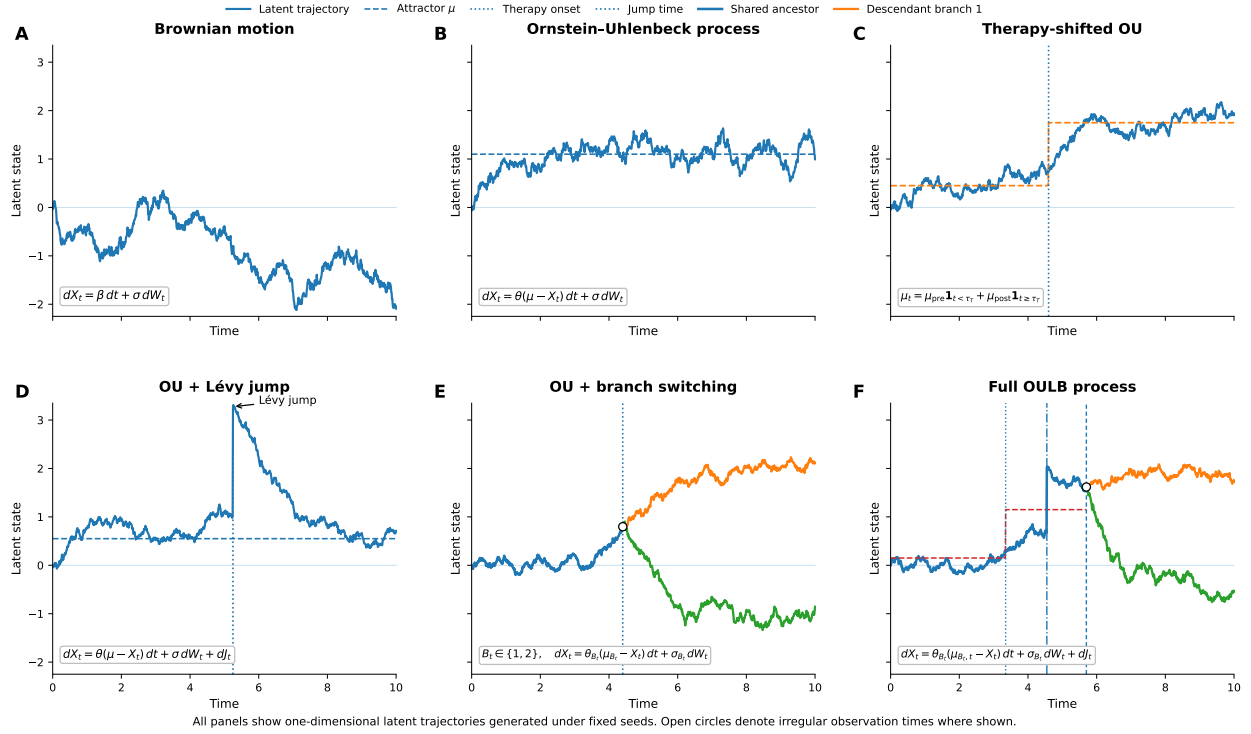
Empirical biological data → modality-specific preprocessing → latent-state representation → empirical observation architecture → OULB calibration and simulation → methodological evaluation and interpretation.
--

This separation between empirical calibration and general stochastic modeling allows OULB to remain biologically interpretable while preserving applicability across diseases, measurement technologies, and biological scales.

Supplementary Figures

The following supplementary figures provide systematic validation of the Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework. Supplementary Figure S1 illustrates the representative stochastic regimes encompassed by the model. Supplementary Figures S2–S4 evaluate recovery of the continuous OU parameters, detection of Lévy jumps, and reconstruction of branch structure, respectively. Supplementary Figure S5 examines joint recovery in a specialized descendant-lineage validation scenario in which continuous dynamics, abrupt jumps, and lineage branching operate simultaneously, whereas Supplementary Figure S6 evaluates robustness to observation noise, irregular sampling, missing data, sample size, latent-state dimensionality, and computational scaling.

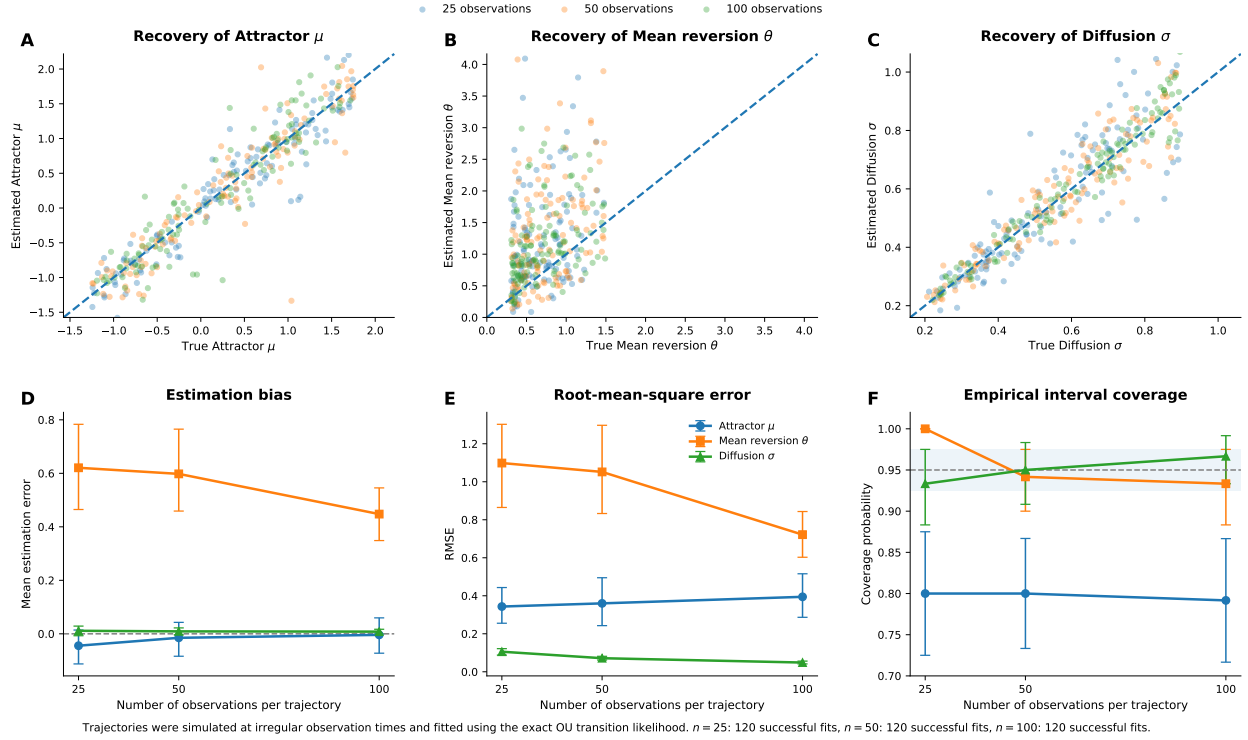
Supplementary Figure S1. Representative simulated stochastic regimes



Supplementary Figure S1. Representative stochastic regimes generated by the OULB framework.

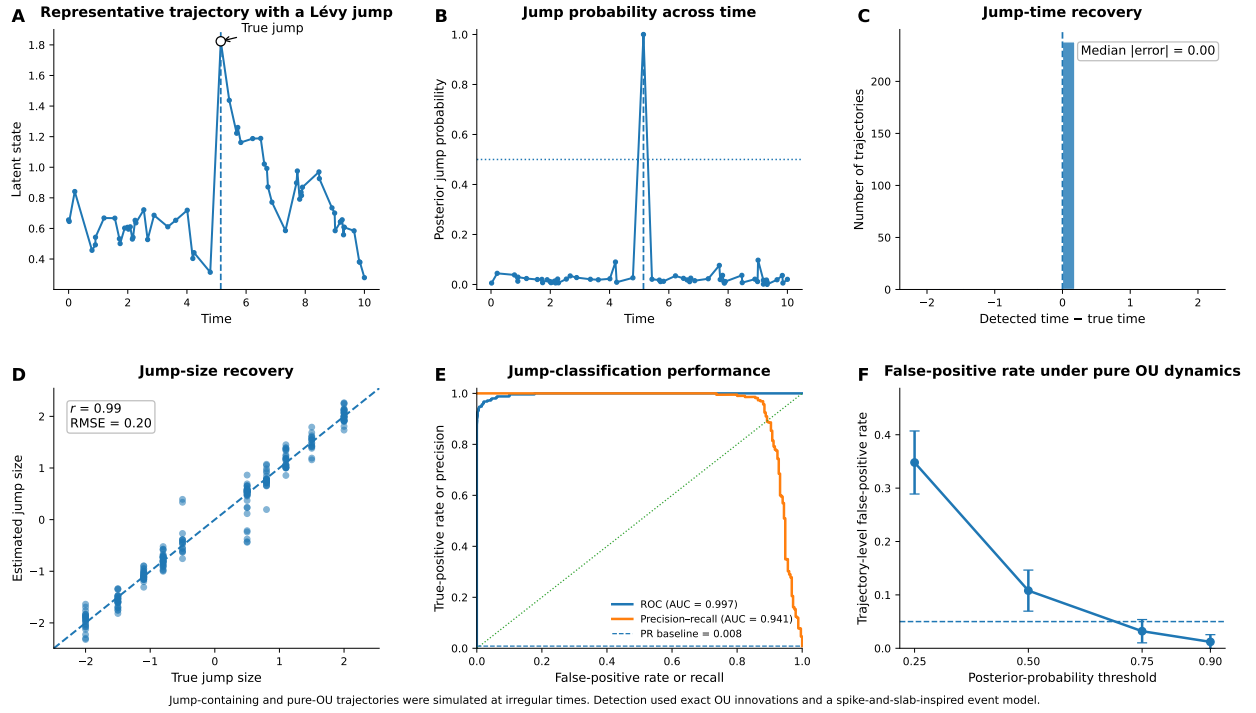
Representative one-dimensional trajectories illustrating the principal stochastic behaviors encompassed by the Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework. (A) Brownian motion exhibiting unconstrained diffusion. (B) Ornstein–Uhlenbeck (OU) dynamics with mean reversion toward a stable attractor. (C) Therapy-dependent OU dynamics following an abrupt attractor shift. (D) OU dynamics with a single Lévy jump. (E) OU dynamics with branch-state switching between lineage-associated attractors. (F) Integrated OULB dynamics combining continuous OU evolution, compound-Poisson jumps, and branch-state switching.

Supplementary Figure S2. Recovery of Ornstein-Uhlenbeck parameters



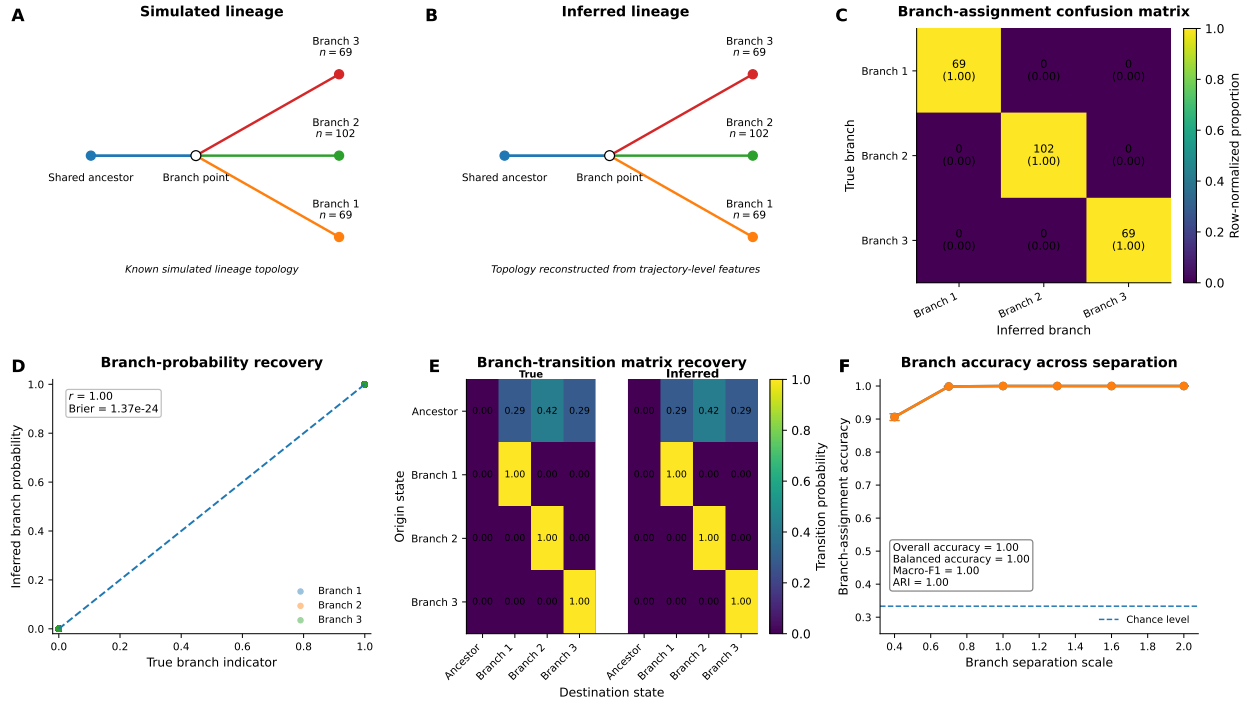
Supplementary Figure S2. Recovery of Ornstein-Uhlenbeck parameters. Simulation-based evaluation of parameter recovery under irregular longitudinal sampling. (A) Recovery of the attractor parameter (μ). (B) Recovery of the mean-reversion parameter (θ). (C) Recovery of the diffusion parameter (σ). (D) Mean estimation bias as a function of the number of observations per trajectory. (E) Root-mean-square error (RMSE) across sampling densities. (F) Empirical 95% interval coverage. Points in panels A–C represent individual simulation replicates, and diagonal dashed lines denote identity. Lines in panels D–F represent mean performance across simulation replicates; error bars denote bootstrap 95% confidence intervals.

Supplementary Figure S3. Validation of Lévy jump detection



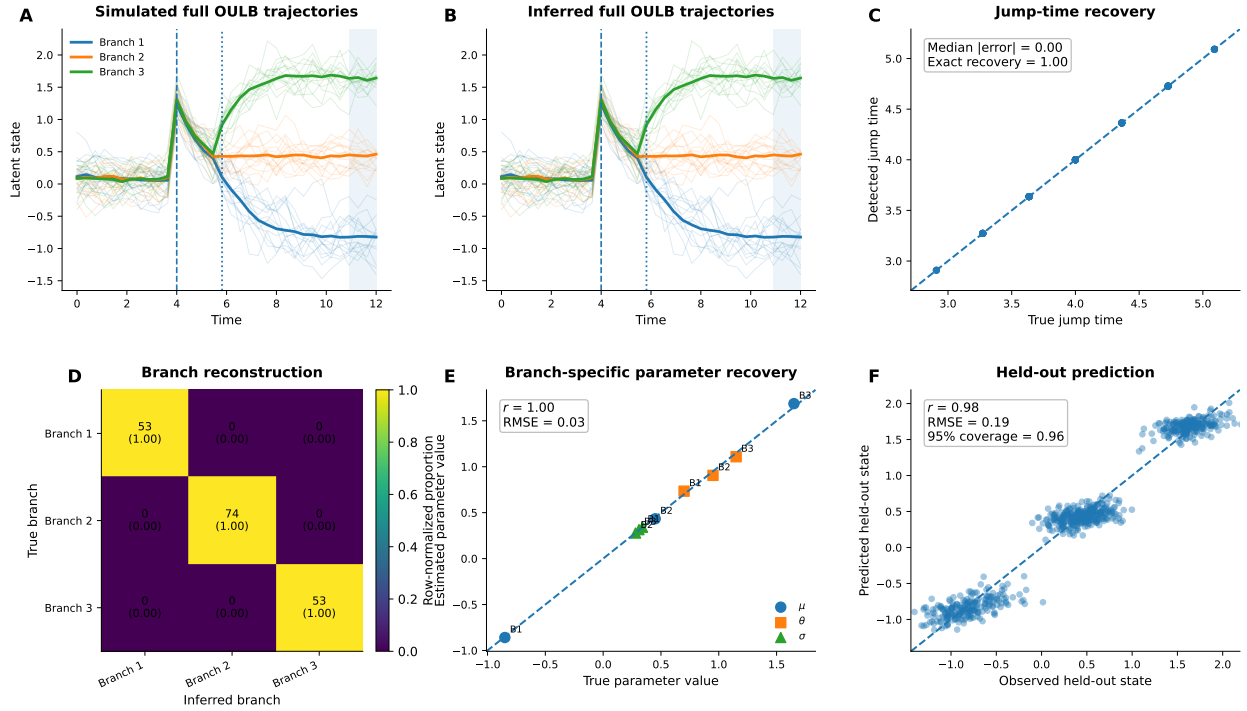
Supplementary Figure S3. Validation of Lévy jump detection. Simulation-based validation of the Lévy jump detection component of the OULB framework. (A) Representative latent trajectory containing a single simulated Lévy jump. (B) Posterior jump probability accurately localizes the discontinuity. (C) Distribution of jump-time recovery errors across simulated trajectories. (D) Recovery of jump size demonstrating strong agreement between true and estimated values. (E) Receiver operating characteristic (ROC) and precision-recall (PR) analyses demonstrating high discrimination performance. (F) Trajectory-level false-positive rate as a function of the posterior jump-probability threshold under pure OU dynamics.

Supplementary Figure S4. Validation of branch reconstruction



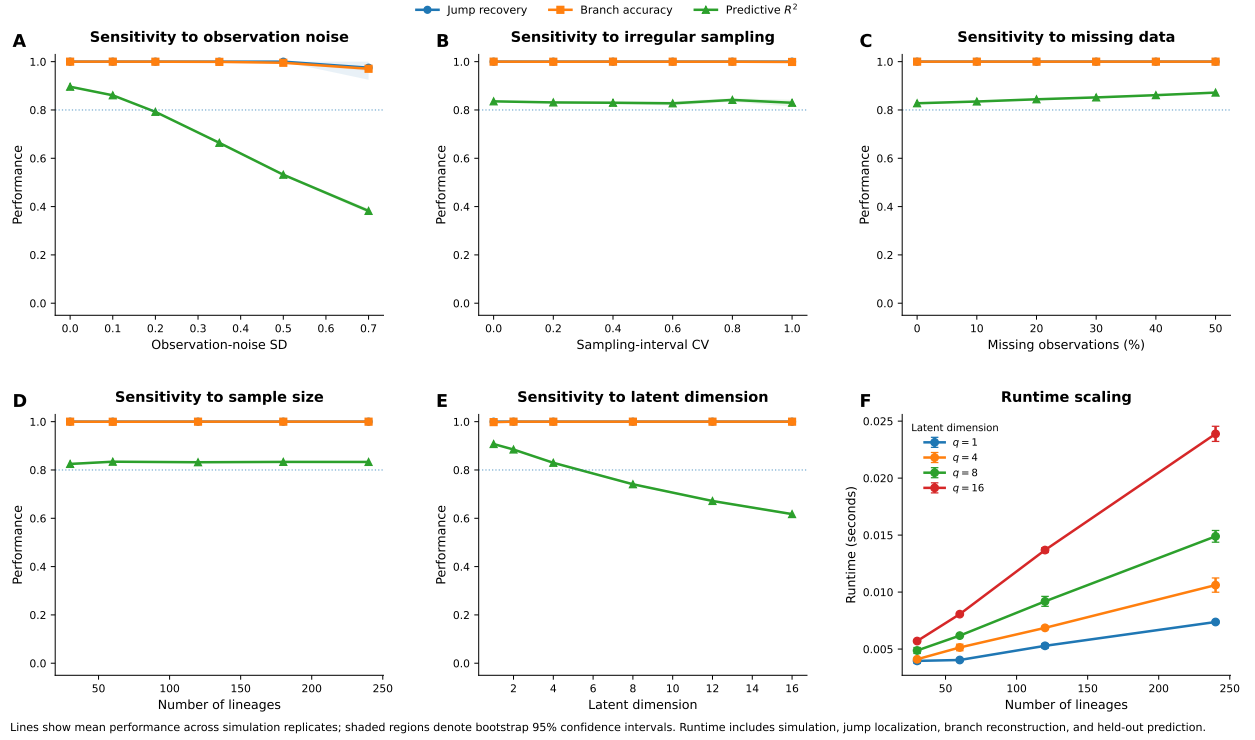
Supplementary Figure S4. Validation of branch reconstruction. Simulation-based validation of lineage reconstruction within the OULB framework. (A) Simulated lineage tree with three descendant branches. (B) Reconstructed lineage inferred from trajectory-level features. (C) Branch-assignment confusion matrix following optimal label alignment. (D) Recovery of branch-membership probabilities relative to the one-hot simulated branch assignments. (E) Comparison of true and inferred branch-transition matrices. (F) Branch-assignment accuracy as a function of branch separation.

Supplementary Figure S5. Joint recovery of the complete OULB model



Supplementary Figure S5. Joint recovery of the complete OULB model. Integrated validation of simultaneous recovery of continuous dynamics, discontinuous events, and descendant-branch structure. (A) Simulated trajectories generated under a specialized joint OULB validation scenario, including continuous OU dynamics, a shared jump event, descendant-lineage branching, and branch-specific stochastic dynamics. (B) Corresponding inferred trajectories. (C) Recovery of the shared jump time. (D) Branch-assignment confusion matrix. (E) Recovery of branch-specific OU parameters (μ , θ , and σ). (F) Prediction of held-out observations using the inferred OULB model.

Supplementary Figure S6. Sensitivity analyses of the OULB framework



Supplementary Figure S6. Sensitivity analyses of the OULB framework. Robustness of the integrated OULB framework under progressively more challenging data conditions. Performance was evaluated with increasing (A) observation noise, (B) irregular sampling, (C) missing observations, (D) varying sample size, and (E) latent-state dimensionality. (F) Computational runtime as a function of sample size and latent dimension. Lines denote mean performance across simulation replicates, and shaded regions represent bootstrap 95% confidence intervals.

Supplementary Tables

The following supplementary tables provide reference information and numerical summaries supporting the mathematical formulation, simulation experiments, and validation analyses. The tables define the notation and prior distributions used in the OULB framework, summarize the simulated stochastic regimes, and report quantitative results for OU parameter recovery, Lévy jump detection, branch reconstruction, joint OULB recovery, and sensitivity analyses.

Supplementary Table S1. Mathematical notation used throughout the OULB framework.

Symbol	Description	Dimension	Units
X_t	Latent biological state at continuous time t	q	latent-state units
Y_i	Observed measurement at observation time t_i	p_i	measurement-specific
q	Dimension of the latent biological state	scalar	—
p_i	Dimension of observation Y_i	scalar	—
B_t	Discrete branch state at time t	discrete, $\{1, \dots, K\}$	—
K	Number of branch states	scalar	—
Θ_b	Branch-specific restoring matrix governing mean reversion	$q \times q$	time ⁻¹
θ	Component-specific restoring parameter used in the diagonal benchmark implementation	scalar or q -vector	time ⁻¹
$\mu_b(t)$	Time-dependent attractor for branch b	q	latent-state units
$\mu_{0,b}$	Baseline or pre-treatment attractor for branch b	q	latent-state units
Δ_b	Branch-specific treatment-induced attractor displacement	q	latent-state units
$u(t)$	Treatment-exposure or external covariate function	scalar or application-specific	application-specific
Σ_b	Branch-specific diffusion matrix	$q \times q$	latent-state units · time ^{-1/2}
σ	Component-specific diffusion scale used in the diagonal benchmark implementation	scalar or q -vector	latent-state units · time ^{-1/2}
W_t	Standard multivariate Wiener process	q	time ^{1/2}
J_t	Compound-Poisson Lévy jump process	q	latent-state units

Continued on next page

Supplementary Table S1. Mathematical notation used throughout the OULB framework (*continued*).

Symbol	Description	Dimension	Units
N_t	Poisson counting process giving the number of jump events up to time t	scalar	count
λ	Compound-Poisson jump intensity	scalar	time^{-1}
Z_k	Jump mark associated with the k th discontinuous event	q	latent-state units
s_J	Per-coordinate jump-scale parameter	scalar	latent-state units
I_q	$q \times q$ identity matrix	$q \times q$	—
Q	Generator matrix of the continuous-time Markov branch-state process	$K \times K$	time^{-1}
$q_{k\ell}$	Transition rate from branch state k to branch state ℓ	scalar	time^{-1}
$P(\Delta)$	Finite-time branch-transition matrix, $P(\Delta) = \exp(Q\Delta)$	$K \times K$	probability
β	Drift vector for the Brownian diffusion baseline	q	latent-state units $\cdot$ time^{-1}
H_i	Observation or projection matrix linking X_{t_i} to Y_i	$p_i \times q$	measurement-specific
ε_i	Observation-error vector	p_i	measurement-specific
R_i	Observation-error covariance matrix	$p_i \times p_i$	measurement units ²
Φ	Collection of OULB model parameters	variable	model-specific
Δ_i	Time interval between consecutive observations, $t_{i+1} - t_i$	scalar	time
$\hat{\theta}_i$	Estimate of a generic parameter in simulation replicate i	scalar	parameter-specific
θ_0	True generating value of a generic parameter in simulation experiments	scalar	parameter-specific

Supplementary Table S2. Prior distributions for Bayesian implementations of the OULB framework.

Parameter or component	Prior	Hyperparameters / interpretation
μ	Normal	$\mu \sim \mathcal{N}(0, 2^2)$; prior for OU attractor location.
θ	Half-normal	$\theta \sim \text{HalfNormal}(1)$; constrains the component-specific restoring parameter to be nonnegative.
σ	Half-normal	$\sigma \sim \text{HalfNormal}(1)$; prior for the component-specific diffusion scale.
Jump intensity λ	Exponential	$\lambda \sim \text{Exponential}(1)$; positive prior for the compound-Poisson jump intensity in fully Bayesian jump-process implementations.
Single-jump amplitude z	Normal	$z \sim \mathcal{N}(0, \tau^2)$; spike-and-slab-inspired prior used for supplementary single-jump event inference where applicable. This quantity is distinct from the compound-Poisson generative jump-scale parameter s_J .
Categorical branch proportions	Dirichlet	$\boldsymbol{\pi} \sim \text{Dirichlet}(1, \dots, 1)$; applies to supplementary probabilistic branch-mixture reconstruction where categorical branch proportions are estimated. It is not a prior on the continuous-time Markov generator Q .
Observation-model parameters	Model dependent	Priors depend on the observation model and measurement scale. For the principal Gaussian observation model, parameters include H_i and R_i ; see Supplementary Note 4.

Notes: The table summarizes priors associated with Bayesian or probabilistic implementations of individual OULB components. Selected simulation benchmarking experiments in the present study used exact-transition maximum-likelihood estimation rather than Bayesian posterior sampling. The compound-Poisson generative jump-mark distribution

$$Z_k \sim \mathcal{N}(0, s_J^2 I_q)$$

specifies the stochastic jump process and should not be interpreted as a prior distribution on s_J . Similarly, the primary branch-state process is parameterized by the continuous-time Markov generator

$$Q = (q_{k\ell}),$$

whereas the Dirichlet prior listed above applies only to categorical branch proportions in specialized supplementary branch-reconstruction analyses.

No universal prior on the entries of Q , the jump-scale parameter s_J , or the observation covariance matrices R_i is imposed by the general OULB architecture; such priors are model- and application-specific in fully Bayesian implementations.

Supplementary Table S3. Simulation scenarios used throughout the validation studies.

Simulation	Purpose	Components activated
Brownian motion	Unconstrained diffusion	Brownian
OU process	Continuous dynamics	OU
Therapy-shifted OU	Dynamic attractor	OU + therapy
OU + Lévy jump	Discontinuous evolution	OU + jump
OU + branch switching	Branch-state dynamics	OU + branch-state process
Full OULB	Complete framework	OU + jump + branching

Supplementary validation experiments were designed independently of the main-figure stress tests and therefore use distinct parameter regimes; numerical values should be interpreted within the simulation settings specified in Supplementary Table S3 and Supplementary Data S1.

Supplementary Table S4. Summary of Ornstein–Uhlenbeck parameter recovery (Supplementary Figure S2).

Parameter	Observations	Mean bias	RMSE	95% coverage
Attractor (μ)	25	−0.044	0.343	0.800
Attractor (μ)	50	−0.015	0.360	0.800
Attractor (μ)	100	−0.004	0.394	0.792
Mean reversion (θ)	25	0.621	1.099	1.000
Mean reversion (θ)	50	0.598	1.052	0.942
Mean reversion (θ)	100	0.448	0.722	0.933
Diffusion (σ)	25	0.011	0.106	0.933
Diffusion (σ)	50	0.009	0.071	0.950
Diffusion (σ)	100	0.008	0.049	0.967

Supplementary Table S5. Summary of Lévy jump detection performance (Supplementary Figure S3).

Metric	Value
Median absolute jump-time error	0.000
Jump-size correlation (r)	0.990
Jump-size RMSE	0.200
Receiver operating characteristic AUC	0.997
Precision–recall AUC	0.941
Precision–recall baseline	0.008

Supplementary Table S6. Summary of branch reconstruction performance (Supplementary Figure S4).

Metric	Value
Overall accuracy	1.000
Balanced accuracy	1.000
Macro-F1 score	1.000
Adjusted Rand index	1.000
Mean Brier score	1.37×10^{-24}

Supplementary Table S7. Joint recovery performance of the complete OULB model (Supplementary Figure S5).

Component	Metric	Value
Jump recovery	Median absolute jump-time error	0.000
Jump recovery	Exact jump-time recovery	1.000
Branch reconstruction	Overall accuracy	1.000
Branch reconstruction	Balanced accuracy	1.000
Branch reconstruction	Macro-F1	1.000
Branch reconstruction	Adjusted Rand index	1.000
Parameter recovery	Correlation	0.999
Parameter recovery	RMSE	0.027
Prediction	Correlation	0.981
Prediction	RMSE	0.191
Prediction	95% interval coverage	0.960

Supplementary Table S8. Summary of sensitivity analyses (Supplementary Figure S6).

Perturbation	Jump recovery range	Branch accuracy range	Predictive R^2 range
Observation noise	0.975–1.000	0.971–1.000	0.382–0.896
Irregular sampling	1.000–1.000	0.999–1.000	0.827–0.842
Missing observations	1.000–1.000	1.000–1.000	0.828–0.872
Sample size	1.000–1.000	1.000–1.000	0.825–0.834
Latent dimension	1.000–1.000	0.998–1.000	0.617–0.907